

The Von Neumann Architecture



**Von Neumann
Architecture**



Machine Language Instructions

- A machine language instruction consists of:
 - Operation code, telling which operation to perform
 - Address field(s), telling the memory addresses of the values on which the operation works.
- Example: ADD X, Y (Add content of memory locations X and Y, and store back in memory location Y).
- Assume: opcode for ADD is 9, and addresses X=99, Y=100

Opcode (8 bits)

Address 1 (16 bits)

Address 2 (16 bits)

00001001	0000000001100011	0000000001100100
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How does this all work together?

- Program Execution:
 - PC is set to the address where the first program instruction is stored in memory.
 - Repeat until HALT instruction or fatal error

Fetch instruction

Decode instruction

Execute instruction

End of loop



Chapter 4

REGISTER TRANSFER AND MICROOPERATIONS



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SIMPLE DIGITAL SYSTEMS

- Combinational and sequential circuits (learned in logic design) can be used to create simple digital systems.
- These are the low-level building blocks of a digital computer.
- Simple digital systems are frequently characterized in terms of
 - the registers they contain, and
 - the operations that they perform.
- Typically,
 - What operations are performed on the data in the registers
 - What information is passed between registers



MICROOPERATIONS

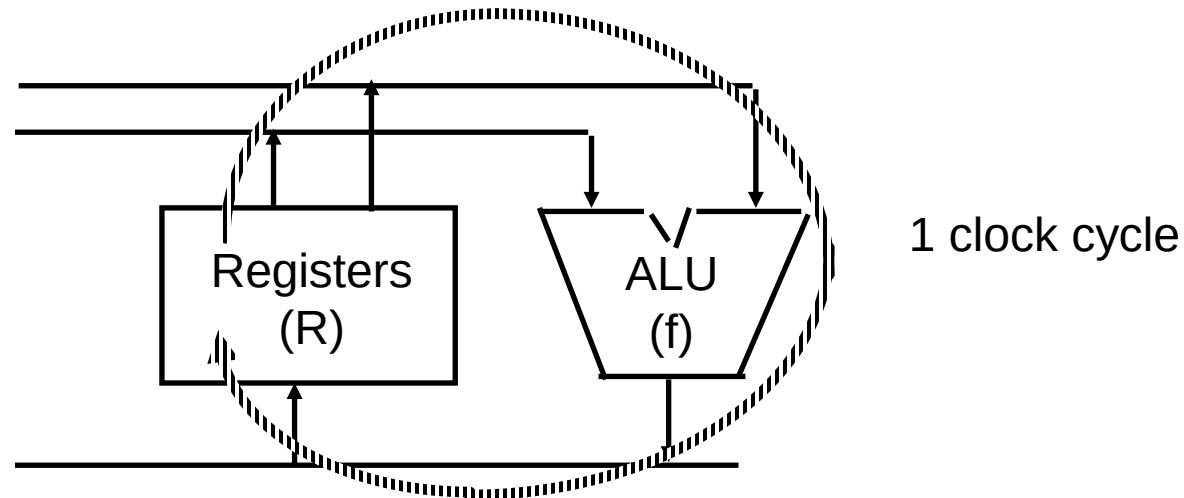
■ Microoperations:

- The operations performed on the data in registers are called microoperations.
- An elementary operation performed on the information stored in one or more registers.
- The functions built into registers are examples of microoperations
 - Shift
 - Load
 - Clear
 - Increment
 - ...



MICROOPERATION

An elementary operation performed (during one clock pulse), on the information stored in one or more registers



$$R \leftarrow f(R, R)$$

f: shift, load, clear, increment, add, subtract, complement, and, or, xor, ...



ORGANIZATION OF A DIGITAL SYSTEM

- Internal organization of a digital computer can be defined by specifying:
 - The set of registers and their functions
 - Microoperations set
 - Set of allowable microoperations provided by the organization of the computer on the binary data in the registers.
 - Control that initiate the sequence of these microoperations (to perform the functions)



REGISTER TRANSFER LEVEL

- Viewing a computer, or any digital system, in this way is called the register transfer level

- This is because we're focusing on
 - The system's registers
 - The data transformations in them, and
 - The data transfers between them.

